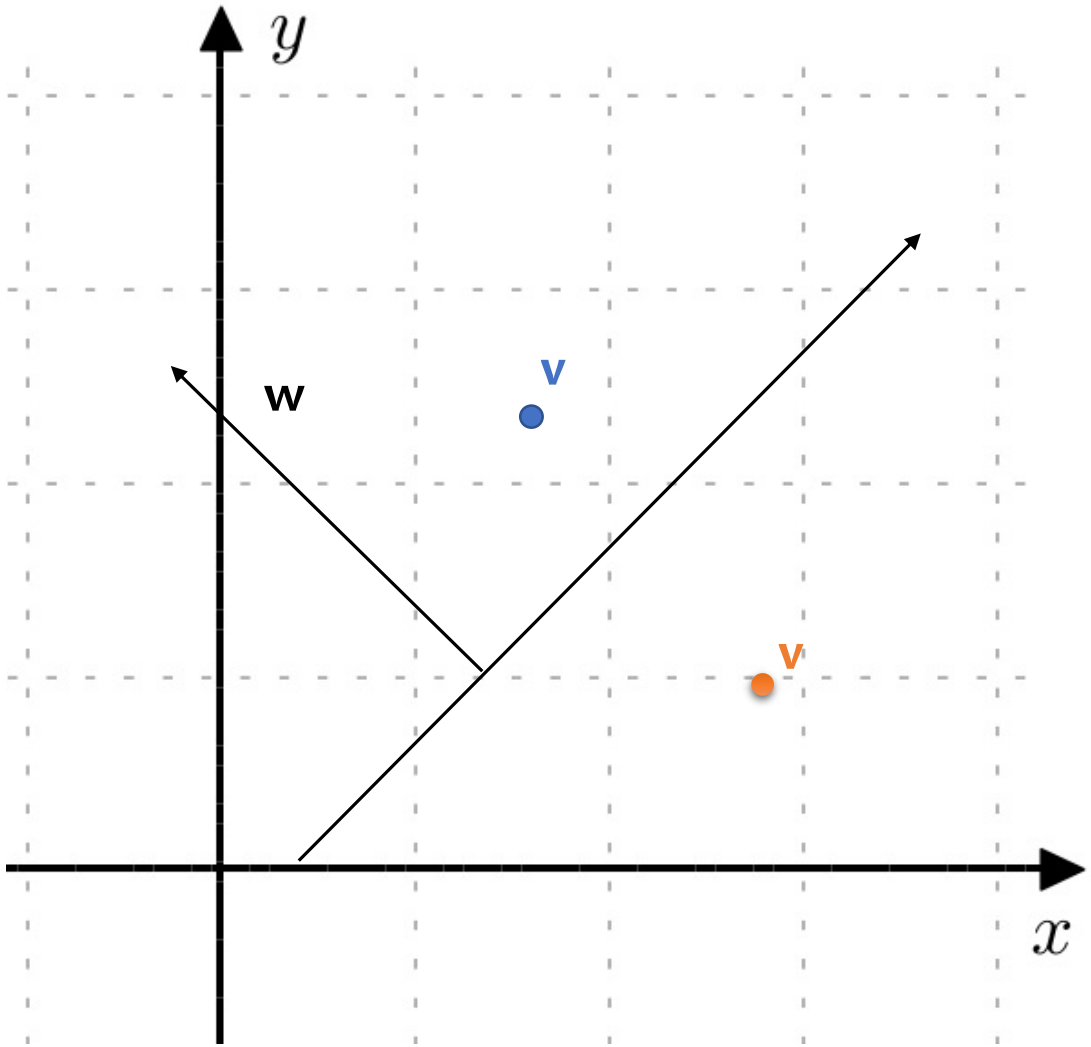
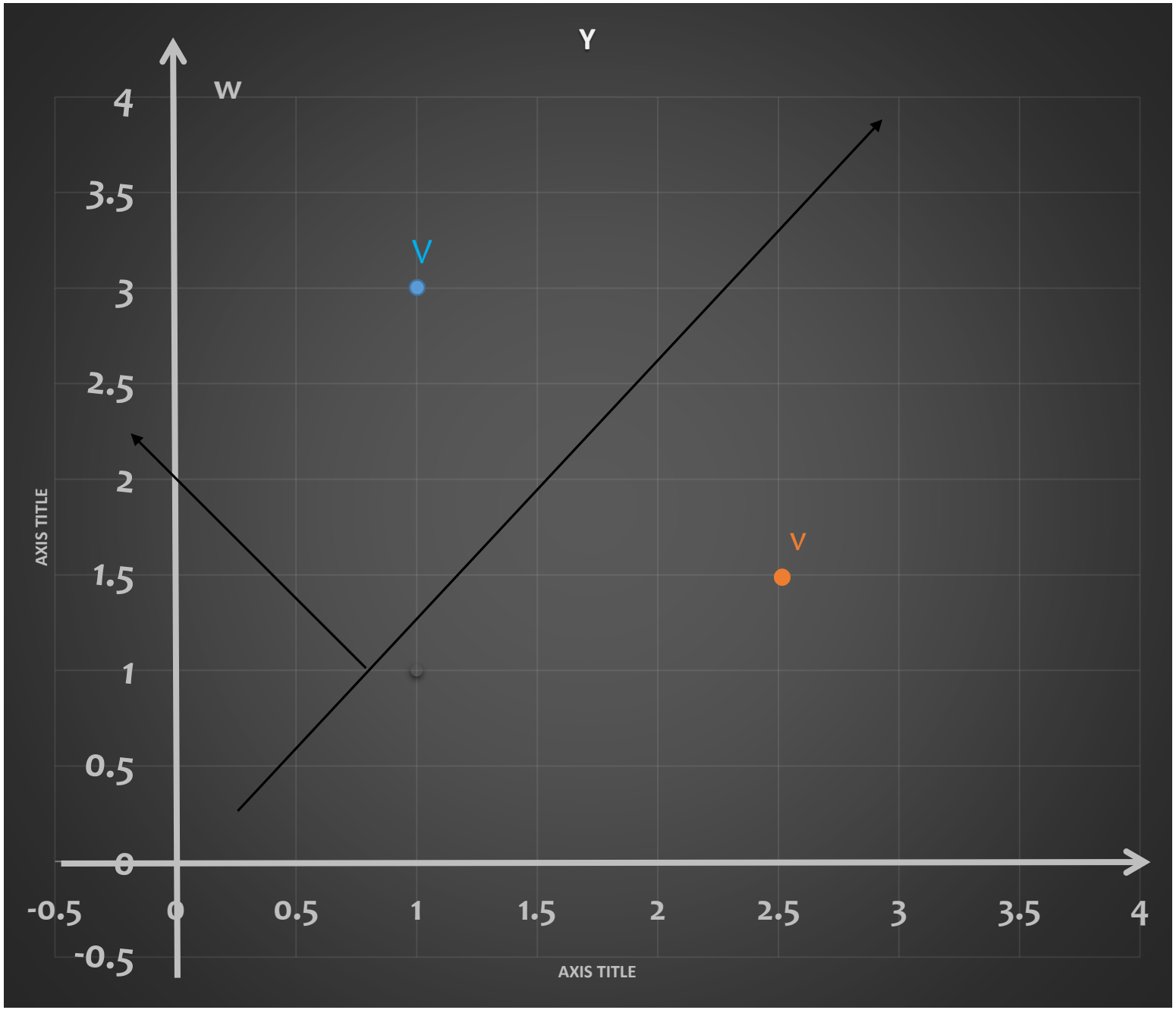
Understanding role of below equation

The Dot Product Formula

$$w.v$$


y
w
v
v
x


Y
w
4
3.5
3
2.5
2
1.5
1
0.5
0
-0.5
V
v
AXIS TITLE
-0.5
0
0.5
1
1.5
2
2.5
3
3.5
4
AXIS TITLE

$$w.v = ||w||\ ||v||\ cos\boldsymbol{\theta}$$

$$||w||\ ||v||\ cos\boldsymbol{\theta} = +d$$

$$||w||\ ||v||\ cos\boldsymbol{\theta} = -d$$

Find our **point v** is below the line or above the line?

$||w|| = \sqrt{x^2 + y^2}$ $||v|| = \sqrt{x^2 + y^2}$

v = $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ w = $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$

Expand $w.v$

$$w.v = ||w|| \; ||v|| \; cos\boldsymbol{\theta}$$

$$||w|| \; ||v|| \; cos\boldsymbol{\theta} = \sqrt{x^2 + y^2} \; \sqrt{x^2 + y^2} \; cos\boldsymbol{\theta}$$

Expand $w.v$

$$w = ||w||$$

$$||w|| = \sqrt{x^2 + y^2}$$

$$v = ||v||$$

$$||v|| = \sqrt{x^2 + y^2}$$

$$w.v = ||w||\ ||v||\ cos\boldsymbol{\theta}$$

$$||w||\ ||v||\ cos\boldsymbol{\theta} = \sqrt{x^2 + y^2}\ \sqrt{x^2 + y^2}\ cos\boldsymbol{\theta}$$

Now Calculate w

w = $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$

w = [x=0,y=2]

$||w|| = \sqrt{x^2 + y^2}$

$||w|| = \sqrt{0^2 + 2^2}$

$||w|| = \sqrt{4}$

||w|| = 1.4142

Calculate v

v = $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

v =[x=1, y=3]

$||v|| = \sqrt{1^2 + 3^2}$

$||v|| = \sqrt{1 + 9}$

$||v|| = \sqrt{10}$

||v|| = 3.17

$||w|| = 1.4142$

$||v|| = 3.17$

$||w||\,||v||\cos(\theta)$

$||w||\,||v||\cos(\theta) = 1.4142 \times 3.17 \times 0.5736$

$||w||\,||v||\cos(\theta) = 4.49 \times 0.5736$

$||w||\,||v||\cos(\theta) = 2.57$

$||w||\,||v||\cos(\theta) = +\,2.57$

Calculate the ||v||

$v = \begin{bmatrix} 2.5 \\ 1.5 \end{bmatrix}$

v =[x= 2.5,y=1.5]

$$||v|| = \sqrt{2.5^2 + 1.5^2}$$

$$||v|| = \sqrt{6.25 + 2.25}$$

$$||v|| = \sqrt{8.5}$$

||v|| = 2.92

$||w|| = 1.4142$

$||v|| = 2.92$

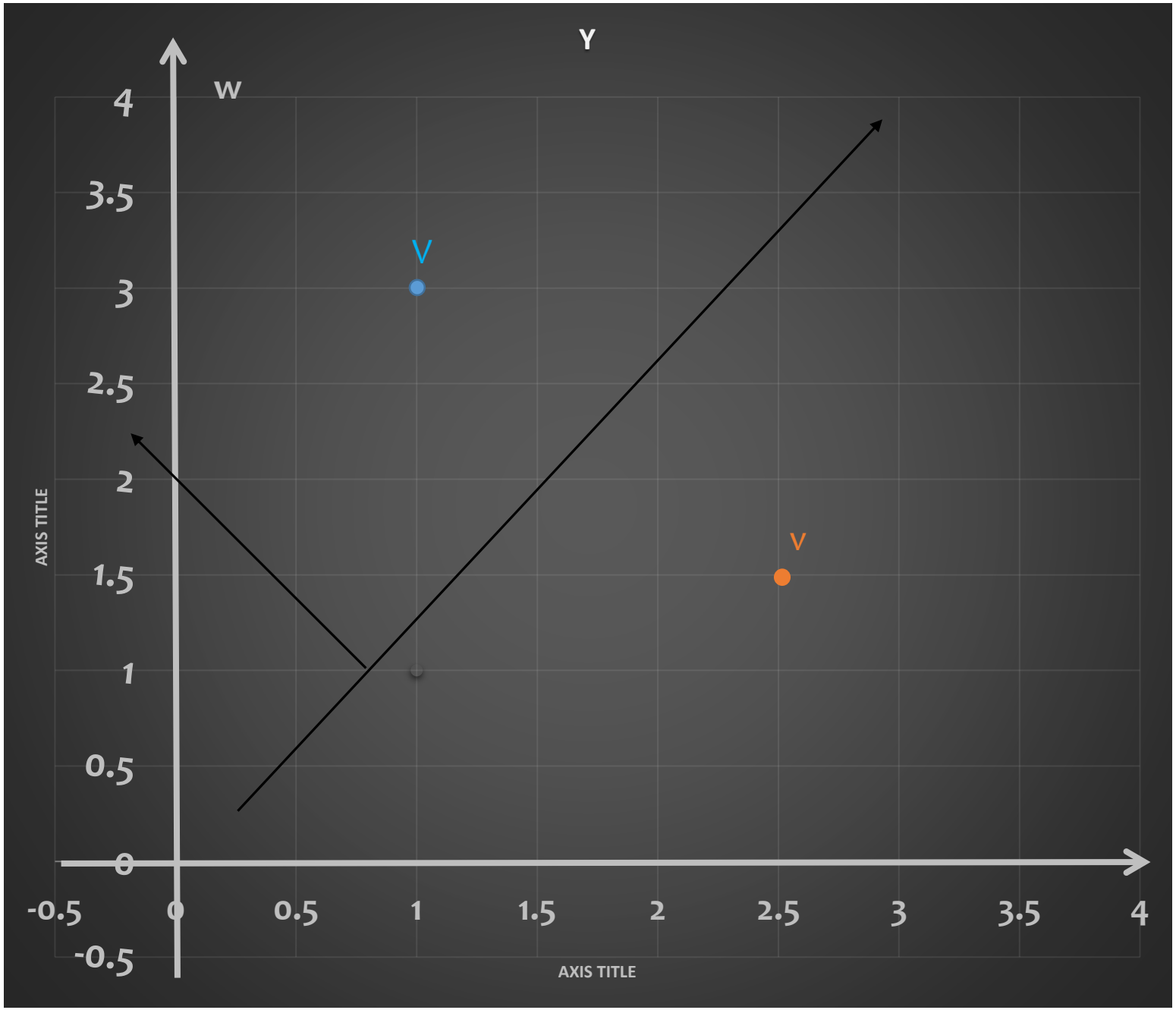
$||w||\ ||x||\ \cos(\theta)$

$||w||\ ||x||\ \cos(\theta) = 1.4142 \times 2.92 \times -0.5000$

$||w||\ ||x||\ \cos(\theta) = 4.129 \times -0.5000$

$||w||\ ||x||\ \cos(\theta) = -2.0645$

$||w||\ ||x||\ \cos(\theta) = -2.0645$


Y
w
4
3.5
3
2.5
2
1.5
1
0.5
0
-0.5
V
v
AXIS TITLE
-0.5
0
0.5
1
1.5
2
2.5
3
3.5
4
AXIS TITLE

$w.v$

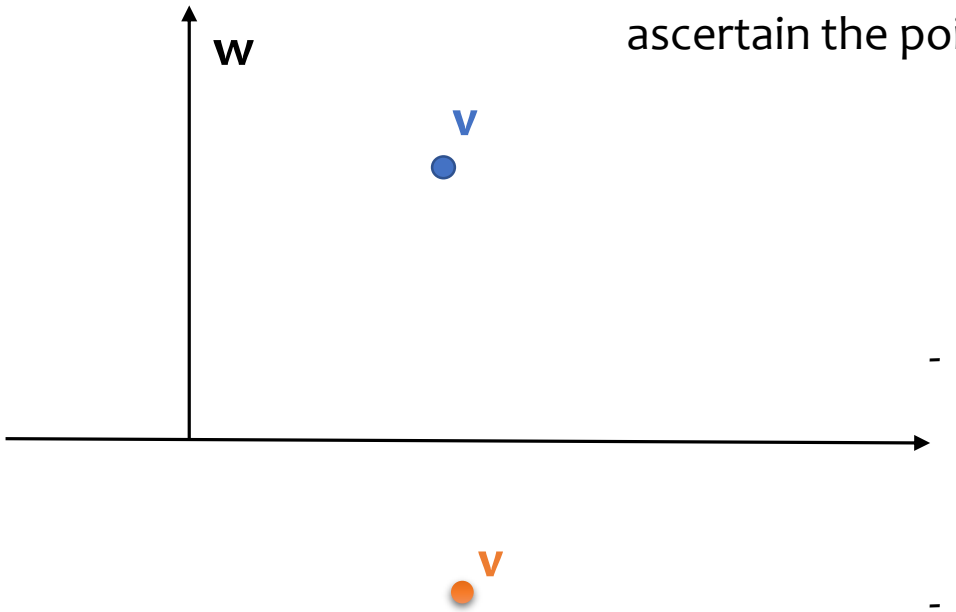
$$||w||\ ||v||\ cos\theta = +d$$

$$||w||\ ||v||\ cos\theta = +d$$

$$||w||\ ||v||\ cos\theta = -d$$

Consider w is a vector which is perpendicular to the plane and v situated above and below the plane respectively. IF point lies on the side of the vector it means it is +ve points and vice versa.

Using the below equation, we can determine the position of the points (values) relative to the plane, weather it lies above or below the hyperplane.



The crucial factor in this equation is the role played by cosine theta (cosθ), as it helps ascertain the point's position based on the angle it forms with the vector.

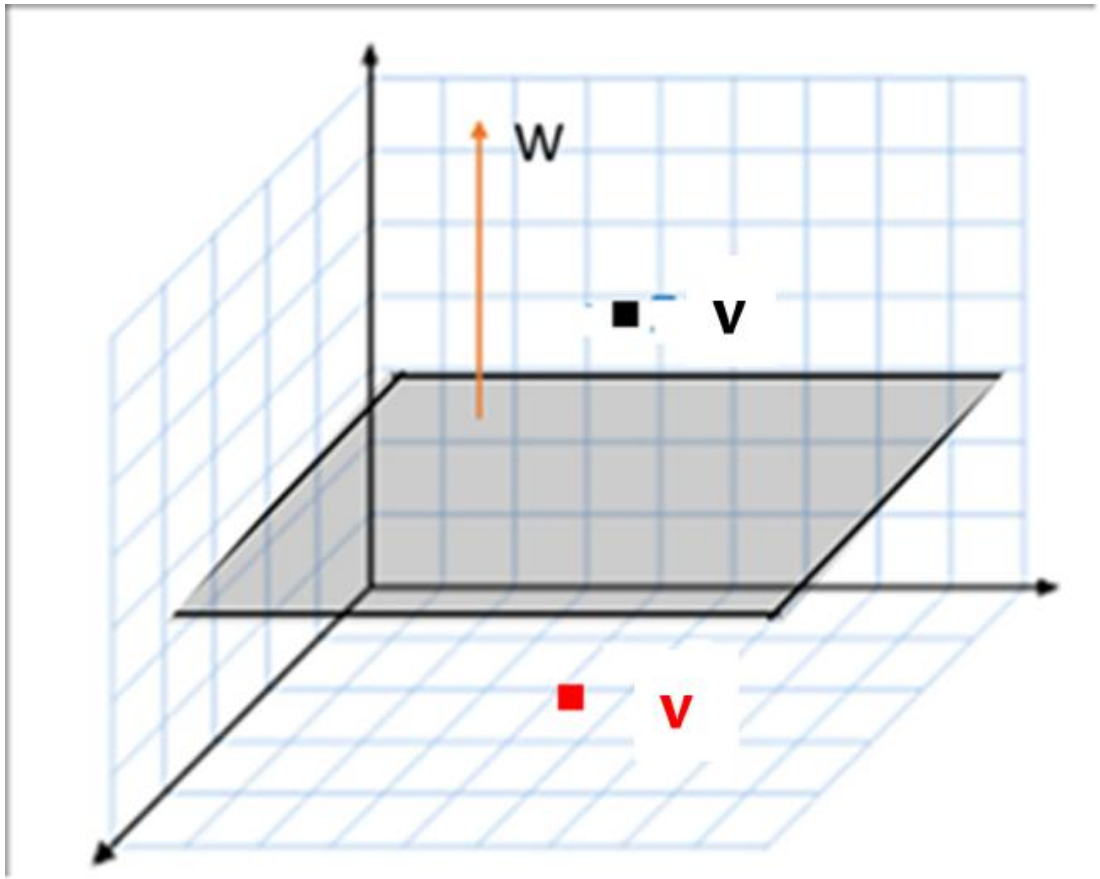
Therefore, when applying the cosθ value in the equation,

- points with an angle **less** than 90 degrees yield a **positive value**,

$$||w||\ ||v||\ cos\boldsymbol{\theta} = +d$$

- while those with an angle **greater** than 90 degrees result in a **negative value**.

$$||w||\ ||v||\ cos\boldsymbol{\theta} = -d$$


w
v
v